Audit Logging

Hostel Management System (HMS) — Software Requirements Specification

Purpose

Define the common audit trail requirements for important actions performed in the Hostel Management System (HMS). Audit records support accountability, troubleshooting, and review of changes to student, hostel, room, bed, application, allotment, payment, complaint, visitor, notice, and account records.

Scope

- Applies to security-sensitive and business-critical events, including sign-in/sign-out, failed sign-in attempts, account and role changes, student-record updates, application decisions, room/bed allocation and release, check-in/check-out, payment recording or status changes, complaint status changes, visitor registration, and administrative configuration changes.

Audit event requirements

- Each event should record a unique event identifier, event timestamp, actor/user identifier when available, actor role, action name, target entity type and identifier, outcome (success/failure), and a concise description.
- Where relevant, capture the request/correlation identifier and source context needed for investigation. Avoid recording passwords, session tokens, authentication secrets, full payment-card data, or unnecessary personal information.
- Record before-and-after values only for fields needed to understand a change. Mask or omit sensitive fields.

Event handling

- Create audit entries through a shared audit-logging component so that services use consistent event names and fields.
- Successful business changes and their audit records should be committed consistently. If an audit record cannot be persisted for a mandatory security event, the system should not silently report the operation as fully audited; apply the configured failure policy and alert an administrator.
- Audit events should be append-only for ordinary application users. Corrections should be represented by a new event rather than editing history.

Access, retention and protection

- Restrict audit-log viewing to authorized administrators or auditors. Apply least-privilege access and log access to sensitive audit data where required.
- Protect logs against unauthorized modification, deletion, and disclosure. Back up audit data with the system's backup process.
- Set retention and archival periods according to institutional policy and applicable requirements. The project must configure the actual retention period before production use.

Review and monitoring

- Provide filters for date range, actor, action, target record, and outcome where audit-log search is implemented.
- Flag repeated authentication failures, permission denials, unusual administrative changes, and repeated failed operations for review.
- Administrators should periodically review security-relevant events and investigate anomalies.

Acceptance criteria

- Critical account and hostel-management actions create an audit event with actor, time, action, target, and outcome.
- Sensitive credentials and tokens are never written to audit logs.
- Ordinary users cannot alter or delete audit history.
- Authorized staff can retrieve relevant events using supported filters.

Document note: This document defines project-level requirements for the HMS utility. Confirm implementation-specific settings (such as role names, file limits, retention periods, and timeout values) against the approved project configuration before production deployment.